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SEPARATOR, PREPARATION METHOD THEREOF, SECONDARY BATTERY, AND ELECTRIC APPARATUS

Abstract

This application provides a separator including a first base film and a second base film, where a melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film. By cooperation of the two layers of base film structures, in a use environment of the separator, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, reducing the risk of short circuits between electrodes and also enhancing the strength of the separator, thereby improving the reliability of the battery in long-term service life.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of International Patent Application No. PCT/CN2023/082347, filed on Mar. 17, 2023, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] This application pertains to the field of secondary battery technologies and specifically relates to a separator, a preparation method thereof, a secondary battery, and an electric apparatus.

BACKGROUND

[0003] Secondary batteries have been widely used in various consumer electronic products and electric vehicles due to their outstanding advantages such as light weight, no pollution, and no memory effect.

[0004] With continuous development of the new energy industry, customers impose higher use requirements on secondary batteries. However, there is a risk of failure caused by short circuit of electrodes during use of batteries. The reliability of secondary batteries is challenged.

SUMMARY

[0005] In view of the technical problems existing in the background, this application provides a separator, a preparation method thereof, a secondary battery, and an electric apparatus, to improve the reliability of the separator and the secondary battery.

[0006] To achieve the above purpose, a first aspect of this application provides a separator including a first base film and a second base film, where a melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film.

[0007] Compared with the prior art, this application includes at least the following beneficial effects: by cooperation of the two layers of base film structures, in a long-term use environment of the separator, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, reducing the risk of short circuits between electrodes and also enhancing the strength of the separator, thereby improving the reliability of the battery in long-term service life.

[0008] In any embodiment of this application, a ratio of the swelling rate of the first base film and the swelling rate of the second base film is less than or equal to 0.8, optionally less than or equal to 0.3.

[0009] When the ratio of the swelling rate of the first base film and the swelling rate of the second base film is within the given ranges, in the long-term use environment of the separator, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, reducing the risk of short circuits between electrodes and also enhancing the strength of the separator, thereby improving the reliability of the battery in long-term service life.

[0010] In any embodiment of this application, the swelling rate of the first base film is smaller than or equal to 5%, optionally 0.2%-2%; and the swelling rate of the second base film is smaller than or

equal to 10%, optionally 1%-8%.

[0011] When the swelling rate of the first base film and the swelling rate of the second base film are respectively within the given ranges, in the long-term use environment of the separator, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, reducing the risk of short circuits between electrodes and also enhancing the strength of the separator, thereby improving the reliability of the battery in long-term service life.

[0012] In any embodiment of this application, a crystallinity of the first base film is smaller than a crystallinity of the second base film.

[0013] In any embodiment of this application, a ratio of the crystallinity of the first base film and the crystallinity of the second base film is 0.4-0.85, optionally 0.4-0.6.

[0014] In any embodiment of this application, the crystallinity of the first base film is 40%-60%, optionally 45%-50%; and the crystallinity of the second base film is 45%-85%, optionally 65%-80%.

[0015] In any embodiment of this application, a branching degree of the first base film is larger than a branching degree of the second base film.

[0016] In any embodiment of this application, the branching degree of the first base film is 0.4-0.9, optionally 0.7-0.9; and the branching degree of the second base film is 0.01-0.4, optionally 0.01-0.2.

[0017] Crystallinity reflects the orderliness of a material structure to some extent. An increase in crystallinity makes the material structure ordered and less prone to swelling, that is, the swelling rate decreases. In contrast to crystallinity that can bring an ordered structure, branching degree reflects an extent to which nonlinear polymer structure branch chains are added to a main chain of the material. A higher branching degree indicates a larger number and higher complexity of side chains formed, making the material more prone to swelling, that is, the swelling rate increases. By adjustment, the crystallinity of the first base film is smaller than the crystallinity of the second base film, the branching degree of the first base film is larger than the branching degree of the second base film, and values are within the above ranges. This can balance the swelling rates of the first base film and the second base film and can enable the first base film and the second base film to have sufficient swelling resistance while having sufficient strength, thereby further improving the reliability of the secondary battery.

[0018] In any embodiment of this application, the melting point of the first base film is 155° C.-365° C., optionally 160° C.-340° C.; and the melting point of the second base film is 130° C.-250° C., optionally 135° C.-220° C.

[0019] When the Celsius melting point of the first base film and the Celsius melting point of the second base film meet the above conditions, the first base film with a higher melting point complements the heat resistance, and the second base film with a lower melting point complements the swelling resistance, so that in a battery environment, the heat resistance of the separator is improved and the swelling degree is reduced, thereby improving the reliability of the battery and prolonging the service life.

[0020] In any embodiment of this application, the first base film and the second base film are each independently selected from at least one of polyolefin and a derivative thereof, halogenated polyolefin and a derivative thereof, polyether and a derivative thereof, polyether ether ketone and a derivative thereof, polyester and a derivative thereof, polyimide and a derivative thereof, polyvinyl alcohol and a derivative thereof, polytetrafluoroethylene and a derivative thereof, polyvinyl fluoride and a derivative thereof, polyvinylidene fluoride and a derivative thereof, and polyethylene terephthalate and a derivative thereof. Using at least one of the above materials as the material for the first base film and the second base film can enable the first base film and the second base film to have good chemical stability and also enable the separator to have low swelling possibility and high strength.

[0021] In any embodiment of this application, a binding layer is further disposed between the first base film and the second base film; the binding layer includes a binder; and optionally, the binding layer includes the binder and a filler. The provision of the binding layer can not only compensate process defects in a base film lamination process, but also further improve the physical properties of the separator, thereby improving the reliability of the secondary battery.

[0022] In any embodiment of this application, the binder includes one or more of polyacrylate, polyacrylic acid, polytetrafluoroethylene, polyvinylidene fluoride, vinylidene fluoride-trichloroethylene copolymer, polyvinylpyrrolidone, polyvinyl acetate, ethylene-vinyl acetate copolymer, polyethylene oxide, polyarylate, carboxymethyl cellulose, hydroxypropyl cellulose, regenerated cellulose, cellulose acetate, cellulose acetate propionate, cellulose acetate butyrate, polyacrylonitrile, polyvinyl alcohol, polyethylene, polypropylene, starch, and cyanoethyl branched starch.

[0023] In any embodiment of this application, the filler includes at least one of inorganic particles, organic particles, and an organic-metal framework material.

[0024] In any embodiment of this application, an air permeability of the separator is less than or equal to 400 s/100 cc, optionally 250 s/100 cc-320 s/100 cc.

[0025] In any embodiment of this application, a transverse direction tensile strength of the separator is 1500 kgf/cm.^{sup.2}-4500 kgf/cm.^{sup.2}, optionally 3000 kgf/cm.^{sup.2}-4500 kgf/cm.^{sup.2}.

[0026] In any embodiment of this application, a longitudinal direction tensile strength of the separator is 1500 kgf/cm.^{sup.2}-4500 kgf/cm.^{sup.2}, optionally 3000 kgf/cm.^{sup.2}-4500 kgf/cm.^{sup.2}.

[0027] In any embodiment of this application, a transverse direction thermal shrinkage rate of the separator at 250° C. for 1 h is less than or equal to 0.4%, optionally less than or equal to 0.2%.

[0028] In any embodiment of this application, a longitudinal direction thermal shrinkage rate of the separator at 250° C. for 1 h is less than or equal to 0.4%, optionally less than or equal to 0.2%.

[0029] When at least one of the above properties of the separator is within the given range, the separator has good physical properties, thereby improving the reliability of the secondary battery.

[0030] A second aspect of this application provides a preparation method of separator. The preparation method includes: providing a first base film and a second base film, where a melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film; and laminating the first base film and the second base film to obtain the separator.

[0031] In any embodiment of this application, the preparation method further includes providing a binding layer slurry, where the binding layer slurry includes a binder; and coating the first base film and/or the second base film with the binding layer slurry to form a binding layer.

[0032] In any embodiment of this application, the binding layer slurry further includes a filler. Optionally, the filler includes at least one of inorganic particles, organic particles, and an organic-metal framework material. When the binding layer slurry further includes the filler and the filler includes at least one of the above materials, the physical properties of the separator can be further improved, thereby improving the reliability of the secondary battery.

[0033] A third aspect of this application provides a secondary battery including the separator according to the first aspect of this application or a separator prepared using the method according to the second aspect of this application. When the secondary battery uses the separator provided, the reliability of the secondary battery can be improved.

[0034] In any embodiment of this application, a positive electrode plate and a negative electrode plate are further included; the separator is disposed between the positive electrode plate and the negative electrode plate; the second base film faces the negative electrode plate, so that the second base film facing the negative electrode plate can complement the swelling resistance, the separator can have more stable physical properties, and in a long-term use environment of the battery, even if

the first base film loses the insulation performance due to swelling, the second base film can complement the performance, preventing short circuits between electrodes and also guaranteeing the strength of the separator, thereby improving the reliability and cycle life of the battery in long-term cycle life.

[0035] A fourth aspect of this application provides an electric apparatus including the secondary battery according to the third aspect of this application. When the secondary battery of the electric apparatus uses the separator provided, the reliability of the electric apparatus can be improved.

[0036] The apparatus of this application includes the secondary battery provided in this application. Therefore, the apparatus includes at least the same advantages as the secondary battery.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0037] To describe the technical solutions of this application more clearly, the following briefly describes the accompanying drawings used in this application. Apparently, the accompanying drawings in the following descriptions show merely some embodiments of this application, and persons of ordinary skill in the art may still derive other drawings from the accompanying drawings without creative efforts.

[0038] FIG. 1 is a schematic structural diagram of an embodiment of a separator of this application.

[0039] FIG. 2 is a schematic structural diagram of an embodiment of a separator of this application.

[0040] FIG. 3 is a schematic flowchart of an embodiment of a preparation method of separator of this application.

[0041] FIG. 4 is a schematic diagram of an embodiment of a secondary battery.

[0042] FIG. 5 is an exploded view of FIG. 4.

[0043] FIG. 6 is a schematic diagram of an embodiment of a battery module.

[0044] FIG. 7 is a schematic diagram of an embodiment of a battery pack.

[0045] FIG. 8 is an exploded view of FIG. 7.

[0046] FIG. 9 is a schematic diagram of an embodiment of an apparatus using a secondary battery as a power source.

DESCRIPTION OF EMBODIMENTS

[0047] The following further describes this application with reference to specific embodiments. It should be understood that the specific embodiments are merely intended to describe this application instead of limiting the scope of this application.

[0048] For brevity, this specification specifically discloses only some numerical ranges. However, any lower limit may be combined with any upper limit to form a range not expressly recorded; any lower limit may be combined with any other lower limit to form a range not expressly recorded; and any upper limit may be combined with any other upper limit to form a range not expressly recorded. In addition, each individually disclosed point or individual single numerical value may itself be a lower limit or an upper limit which can be combined with any other point or individual numerical value or combined with another lower limit or upper limit to form a range not expressly recorded.

[0049] In the description of this specification, unless otherwise specified, the term “or (or)” is inclusive. That is, the phrase “A or (or) B” means “A, B, or both A and B”. More specifically, any one of the following conditions satisfies the condition “A or B”: A is true (or present) and B is false (or not present); A is false (or not present) and B is true (or present); or both A and B are true (or present).

[0050] In the description of this specification, it should be noted that “more than” and “less than” are inclusive of the present number and that “more” in “one or more” means two or more than two, unless otherwise specified.

[0051] Unless otherwise specified, the terms used in this application have well-known meanings as commonly understood by persons skilled in the art. Unless otherwise specified, numerical values of parameters mentioned in this application may be measured by using various measurement methods commonly used in the art (for example, they may be tested by using the methods provided in the embodiments of this application).

Secondary Battery

[0052] A secondary battery refers to a battery that can be charged after being discharged to activate active materials for continuous use.

[0053] Generally, a secondary battery includes a positive electrode plate, a negative electrode plate, a separator, and an electrolyte. During charging and discharging of the battery, active ions are intercalated and deintercalated back and forth between the positive electrode plate and the negative electrode plate. The separator is disposed between the positive electrode plate and the negative electrode plate for separation. The electrolyte conducts ions between the positive electrode plate and the negative electrode plate.

[Separator]

[0054] Referring to FIG. 1, an embodiment of this application provides a separator **10** including a first base film **11** and a second base film **12**, where a melting point of the second base film **12** is lower than a melting point of the first base film **11**, and a swelling rate of the first base film **11** is smaller than a swelling rate of the second base film **12**. By cooperation of the two layers of base film structures, the separator can have both good heat resistance and good reliability.

[0055] Without being limited by any theory, in the specific separator structure of this application, the base films with different swelling rates are laminated to make the separator have more stable physical properties, so that in a long-term use environment of the battery, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, preventing short circuits between electrodes and also guaranteeing the strength of the separator, thereby improving the reliability and cycle life of the battery in long-term cycle life.

[0056] In any embodiment of this application, a ratio of the swelling rate of the first base film and the swelling rate of the second base film is less than or equal to 0.8, optionally less than or equal to 0.3. For example, the ratio of the swelling rate of the first base film and the swelling rate of the second base film is optionally 0.001, 0.01, 0.03, 0.05, 0.06, 0.08, 0.1, 0.15, 0.2, 0.25, 0.31, 0.35, 0.4, 0.45, 0.5, 0.6, 0.65, 0.7, 0.8, or the like, or a range defined by any two of these values. For example, the ratio of the swelling rate of the first base film to the swelling rate of the second base film is optionally 0.001-0.01, 0.01-0.08, 0.02-0.06, 0.04-0.08, 0.1-0.15, 0.2-0.25, 0.25-0.35, 0.35-0.45, 0.5-0.65, 0.65-0.8, or the like.

[0057] When the ratio of the swelling rate of the first base film and the swelling rate of the second base film is within the given ranges, in the long-term use environment of the separator, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, preventing short circuits between electrodes and also enhancing the strength of the separator, thereby improving the reliability and cycle life of the battery in long-term service life.

[0058] In any embodiment of this application, the swelling rate of the first base film is less than or equal to 5%, optionally 0.2%-2%. For example, the swelling rate of the first base film is optionally 0.1%, 0.3%, 0.5%, 0.7%, 1.1%, 1.4%, 1.8%, 2.2%, 2.4%, 2.9%, 3.4%, 3.6%, 4.1%, 4.8%, 5%, or the like, or a range defined by any two of these values. For example, the swelling rate of the first base film is optionally 0.01%-0.1%, 0.1%-5%, 0.1%-3.4%, 0.1%-2.9%, 0.1%-2.2%, 0.1%-0.5%, 0.5%-5%, 0.5%-4.1%, 0.5%-3.4%, 0.5%-2.4%, 0.5%-1.1%, 1.1%-5%, 1.1%-4.1%, 1.1%-3.6%, 1.1%-3.4%, 1.1%-1.8%, 2.2%-5%, 2.2%-4.8%, 2.2%-4.1%, 2.2%-3.6%, 2.2%-2.9%, 3.4%-5%, 3.4%-4.1%, 4.1%-5%, or the like.

[0059] In any embodiment of this application, the swelling rate of the second base film is less than or equal to 10%, optionally 1%-8%. For example, the swelling rate of the second base film is

optionally 0.2%, 0.4%, 0.7%, 0.9%, 1.0%, 1.2%, 1.4%, 1.6%, 1.8%, 1.9%, 2.0%, 2.1%, 2.6%, 3.5%, 3.7%, 4.4%, 6.4%, 7.8%, 9.2%, 10%, or the like, or a range defined by any two of these values. For example, the swelling rate of the second base film is optionally 0.01%-0.05%, 0.05%-0.2%, 0.05%-1.0%, 0.05%-1.5%, 0.05%-2.0%, 0.05%-5.0%, 0.05%-8.0%, 0.3%-0.5%, 1.0%-1.5%, 1.0%-2.0%, 1.5%-5.0%, 2.0%-8.0%, 5.0%-8.0%, 3.0%-8.0%, 7.5%-9.0%, 0.2%-0.7%, 0.7%-1.0%, 1.0%-1.6%, 1.8%-2.0%, 2.1%-2.6%, 2.6%-3.7%, 4.4%-6.4%, 7.8%-10%, or the like.

[0060] When the ratio of the swelling rate of the first base film and the swelling rate of the second base film is within the given ranges, the reliability of the battery in long-term cycle life can be further improved.

[0061] The swelling rate of the base film has a meaning well known in the art, and can be measured using methods known in the art. For example, the following testing method may be used: six groups of base film parallel samples are taken, with a size of 50 mm×100 mm; the samples are soaked in an electrolyte, totally immersed in the electrolyte, and placed in an environment at 70° C.; and mass changes are monitored every 24 hours for seven consecutive days. A mass increasing rate of the base film is the swelling rate. A solvent used in the electrolyte includes ethylene carbonate (EC) and ethyl methyl carbonate (EMC) at a mass ratio of 30:70, LiPF₆ is used as an electrolytic salt, and a concentration of the electrolytic salt is 1.0 mol/L.

[0062] Persons skilled in the art know that the swelling rate of the base film can be adjusted by adjusting intrinsic parameters (for example, one or more of the crystallinity, the branching degree, and the like of the material of the base film) of the base film and preparation process parameters (for example, one or more of a stretch ratio, a stretch temperature, and the like) of the base film. For example, under a condition that the other conditions remain unchanged, a larger crystallinity of the base film results in a smaller swelling rate, and a larger branching degree of the base film results in a larger swelling rate. Persons skilled in the art can adjust the swelling rate of the base film within the parameter ranges provided in this application using known methods. For example, a base film with a desired swelling rate can be achieved through a limited number of experiments by adjusting a production process of the base film (adjusting the crystallinity, the branching degree, and the like).

[0063] In any embodiment of this application, the crystallinity of the first base film is smaller than the crystallinity of the second base film.

[0064] In some optional embodiments, the ratio of the crystallinity of the first base film and the crystallinity of the second base film is 0.4-0.85, preferably 0.4-0.6. For example, the ratio of the crystallinity of the first base film and the crystallinity of the second base film is optionally 0.4, 0.5, 0.6, 0.7, 0.8, 0.85, or the like, or a range defined by any two of these values. For example, the ratio of the crystallinity of the first base film and the crystallinity of the second base film is optionally 0.4-0.5, 0.5-0.7, 0.7-0.85, or the like.

[0065] In some optional embodiments, the crystallinity of the first base film is optionally 40%-60%, optionally 45%-50%; and the crystallinity of the first base film is specifically optionally 40%, 42%, 46%, 49%, 51%, 52%, 54%, 57%, 59%, 60%, or the like, or a range defined by any two of these values. For example, the crystallinity of the first base film is specifically optionally 40%-42%, 42%-49%, 49%-52%, 52%-57%, 57%-60%, or the like. The crystallinity of the second base film is optionally 45%-85%, optionally 65%-80%. The crystallinity of the second base film is specifically optionally 45%, 46%, 49%, 54%, 61%, 66%, 74%, 77%, 81%, 84%, 85%, or the like, or a range defined by any two of these values. For example, the crystallinity of the second base film is specifically optionally 45%-46%, 46%-54%, 54%-66%, 74%-77%, 81%-85%, or the like.

[0066] Crystallinity reflects orderliness of the material structure to some extent. An increase in crystallinity makes the material structure ordered and less prone to swelling, that is, the swelling rate decreases. The crystallinity of the first base film and the crystallinity of the second base film are adjusted to be respectively within the above ranges, so that the swelling rates of the first base film and the second base film can be adjusted, allowing the first base film and the second base film

to complement each other better in performance. Furthermore, the orderliness of the material structure also affects the strength of the material. The crystallinity of the first base film and the crystallinity of the second base film are adjusted to be within the above ranges, so that the first base film and the second base film can have good strength on the premise of meeting the swelling requirement of this application, thereby improving the reliability of the secondary battery in the long-term service life.

[0067] The crystallinity of the base film has a meaning well known in the art and can be tested using devices and methods known in the art. For example, the crystallinity of the base film is determined using differential scanning calorimetry (DSC method). Specifically, the following testing method may be used: 4-6 mg of a to-be-tested base film sample is placed in a sample chamber of a differential scanning calorimeter and heated from 25° C. to 350° C. at a heating rate of 10° C./min to obtain a melting endothermic curve, and a peak area of the curve and a reference value of 100% crystalline polyolefin are calculated to obtain the crystallinity.

[0068] In any embodiment of this application, the branching degree of the first base film is larger than the branching degree of the second base film.

[0069] In any embodiment of this application, the branching degree of the first base film is 0.4-0.9, optionally 0.7-0.9; and the branching degree of the second base film is 0.01-0.4, optionally 0.01-0.2. For example, the branching degree of the first base film may be 0.4, 0.5, 0.6, 0.7, 0.75, 0.78, 0.8, 0.9, or a range defined by any two of these values. For example, the branching degree of the first base film is 0.4-0.6, 0.6-0.75, 0.75-0.78, 0.78-0.9. The branching degree of the second base film is optionally 0.01, 0.03, 0.09, 0.14, 0.18, 0.25, 0.33, 0.38, 0.4, or the like, or a range defined by any two of these values. For example, the branching degree of the second base film is optionally 0.01-0.03, 0.03-0.14, 0.14-0.25, 0.25-0.4, or the like.

[0070] In contrary to crystallinity that can bring an ordered structure, branching degree refers to an extent to which non-linear polymer structure branch chains are added onto a main chain of the base film material, meaning that a higher branching degree indicates a larger number and higher complexity of side chains formed. A higher branching degree indicates a larger swelling rate. The ratio of the branching degree of the first base film and the branching degree of the second base film are selected within the above ranges, so that the swelling rates of the first base film and the second base film can be adjusted, allowing the first base film and the second base film to complement each other better in performance. Further, crystallinity also affects the swelling rate of the material. The branching degree and the crystallinity are jointly adjusted to be within the above ranges, so that the first base film and the second base film can have good strength on the premise of meeting the swelling requirement of this application, thereby improving the reliability of the secondary battery in the long-term service life.

[0071] The branching degree of the base film has a meaning well known in the art and can be tested using devices and methods known in the art. For example, infrared spectroscopy may be used. Specifically, the operation may be performed as follows: 3-5 groups of base film samples are selected, and the samples are placed in an infrared spectrometer for scanning, within a scan range of 400-4000 cm.sup.-1. After scanning is completed, an infrared curve of the samples is processed to calculate a ratio of a peak area at 1377 cm.sup.-1 and a peak area at 1996-2062 cm.sup.-1, where the ratio is the branching degree.

[0072] In some embodiments, the first base film satisfies both that the crystallinity is 45%-50% and the branching degree is 0.7-0.9.

[0073] In some embodiments, the second base film satisfies both that the crystallinity is 65%-80% and the branching degree is 0.01-0.2.

[0074] When the crystallinity and branching degree of the first base film and/or the second base film are both within the given ranges, the two base films can have good strength performance while achieving the swelling rate designed in this application.

[0075] In any embodiment of this application, the melting point of the first base film is 155°

C.-360° C., optionally 160° C.-340° C.; and the melting point of the second base film is 130° C.-250° C., optionally 130° C.-220° C. For example, the melting point of the first base film is optionally 155° C., 165° C., 175° C., 188° C., 191° C., 220° C., 239° C., 245° C., 271° C., 294° C., 310° C., 324° C., 347° C., 360° C., or the like, or a range defined by any two of these values. For example, the melting point of the first base film is optionally 155° C.-175° C., 175° C.-220° C., 220° C.-245° C., 245° C.-271° C., 271° C.-310° C., 310° C.-360° C., or the like. For example, the melting point of the second base film is optionally 130° C., 131° C., 133° C., 136° C., 145° C., 151° C., 163° C., 172° C., 174° C., 210° C., 225° C., 236° C., 250° C., or the like, or a range defined by any two of these values. For example, the melting point of the second base film is optionally 130° C.-131° C., 131° C.-136° C., 145° C.-151° C., 163° C.-174° C., 174° C.-210° C., 210° C.-250° C., or the like.

[0076] The Celsius melting point of the first base film and the Celsius melting point of the second base film are adjusted to meet the above conditions, so that the first base film with a higher melting point complements the heat resistance, and the second base film with a lower melting point complements the swelling resistance, allowing the separator to have good heat resistance and swelling resistance.

[0077] According to some embodiments, the melting points of the first base film and the second base film have meanings well known in the art and can be measured using methods known in the art. For example, differential scanning calorimetry can be used for measurement. For specific details, reference can be made to the standard GB/T 19466.3-2004. In an example, the measurement can be performed according to the following method: 4-6 mg of a to-be-tested sample is placed in a sample chamber of a differential scanning calorimeter and heated from 25° C. to 400° C. at a heating rate of 10° C./min, to obtain a melting endothermic curve of the sample, and a temperature corresponding to a peak of the curve is a melting point of the sample.

[0078] In any embodiment of this application, the first base film and the second base film are each be independently made of a material selected from at least one of polyolefin and a derivative thereof, halogenated polyolefin and a derivative thereof, polyether and a derivative thereof, polyether ether ketone and a derivative thereof, polyester and a derivative thereof, polyimide and a derivative thereof, polyvinyl alcohol and a derivative thereof, polytetrafluoroethylene and a derivative thereof, polyvinyl fluoride and a derivative thereof, polyvinylidene fluoride and a derivative thereof, and polyethylene terephthalate and a derivative thereof. For example, the first base film may be made of polyolefin and a derivative thereof or halogenated polyolefin and a derivative thereof. The first base film may alternatively be made of polyether and a derivative thereof, polyether ether ketone and a derivative thereof, polyester and a derivative thereof, or the like. The first base film and the second base film may alternatively be made of polyvinylidene fluoride and a derivative thereof, polyethylene terephthalate, or the like. The derivatives generally refer to products derived when hydrogen atoms or atomic groups in compounds are substituted by other atoms or atomic groups. Using at least one of the above materials as the material for the first base film and the second base film can improve the chemical stability of the first base film and the second base film and also enable the separator to have low swelling possibility and high strength.

[0079] Referring to FIG. 2, an embodiment of this application provides a separator **10** including a first base film **11** and a second base film **12**. In any embodiment of this application, a binding layer **13** is further disposed between the first base film **11** and the second base film **12**, where the binding layer includes a binder. Optionally, the binding layer includes the binder and a filler.

[0080] When the binding layer is disposed between the first base film and the second base film and the binding layer includes the binder, process defects in a hot-pressing lamination process of the base film can be compensated, and the physical properties (such as the tensile strength, the puncture strength, and the heat resistance) of the separator can also be further improved. Thus, the reliability of the secondary battery is improved. The filler is located between the first base film and the second base film, and therefore can reduce the risk of powder falling.

[0081] In any embodiment of this application, the binder includes one or more of polyacrylate, polyacrylic acid, polytetrafluoroethylene, polyvinylidene fluoride, vinylidene fluoride-trichloroethylene copolymer, polyvinylpyrrolidone, polyvinyl acetate, ethylene-vinyl acetate copolymer, polyethylene oxide, polyarylate, carboxymethyl cellulose, hydroxypropyl cellulose, regenerated cellulose, cellulose acetate, cellulose acetate propionate, cellulose acetate butyrate, polyacrylonitrile, polyvinyl alcohol, polyethylene, polypropylene, starch, and cyanoethyl branched starch.

[0082] In any embodiment of this application, the filler includes at least one of inorganic particles, organic particles, and an organic-metal framework material.

[0083] Optionally, the inorganic particles include one or more of inorganic particles with a dielectric constant above 5, inorganic particles with ionic conductivity but not storing ions, and inorganic particles capable of having electrochemical reactions.

[0084] Optionally, the inorganic particles with a dielectric constant above 5 may include at least one of boehmite, aluminum oxide, zinc oxide, silicon dioxide, titanium oxide, zirconium oxide, barium oxide, calcium oxide, magnesium oxide, nickel oxide, tin oxide, cerium oxide, yttrium oxide, hafnium oxide, aluminum hydroxide, magnesium hydroxide, silicon carbide, boron carbide, aluminum nitride, silicon nitride, boron nitride, magnesium fluoride, calcium fluoride, barium fluoride, barium sulfate, magnesium aluminum silicate, magnesium lithium silicate, magnesium sodium silicate, bentonite, hectorite, zirconium titanate, barium titanate, $\text{Pb}(\text{Zr,Ti})\text{O}_{0.3}$ (PZT for short), $\text{Pb}_{1-m}\text{La}_m\text{Zr}_{1-n}\text{Ti}_n\text{O}_{0.3}$ (PLZT for short, $0 < m < 1$, and $0 < n < 1$),

$\text{Pb}(\text{Mg}_{0.3}\text{Nb}_{0.2/3})\text{O}_{0.3}$ — $\text{PbTiO}_{0.3}$ (PMN-PT for short), and respective modified inorganic particles thereof. Optionally, the inorganic particles may be modified through chemical modification and/or physical modification. The chemical modification includes modification by a coupling agent (for example, a silane coupling agent, a titanate coupling agent, or the like), modification by a surfactant, graft modification by a polymer, and the like. The physical modification may include mechanical force dispersion, ultrasonic dispersion, high-energy treatment, and the like. Agglomeration of inorganic particles can be reduced through the modification treatment, so that the binding layer can have a more stable and more uniform structure. In addition, selecting a coupling agent, surfactant, or polymer with a specific functional group to modify inorganic particles also helps to improve the wettability and retention characteristics of the binding layer to the electrolyte and improve the adhesion performance of the binding layer to the first base film and the second base film.

[0085] Optionally, the inorganic particles with ionic conductivity but not storing ions may include at least one of Li_3PO_4 , lithium titanium phosphate $\text{Li}_{x1}\text{Ti}_{y1}(\text{PO}_4)_{0.3}$, lithium aluminum titanium phosphate $\text{Li}_{x2}\text{Al}_{y2}\text{Ti}_{z1}(\text{PO}_4)_{0.3}$, $(\text{LiAlTiP})_{x3}\text{O}_{y3}$ type glass, lithium lanthanum titanate $\text{Li}_{x4}\text{La}_{y4}\text{TiO}_{0.3}$, lithium germanium thio-phosphate $\text{Li}_{x5}\text{Ge}_{y5}\text{P}_{z2}\text{S}_w$, lithium nitride $\text{Li}_{x6}\text{N}_{y6}$, SiS_2 type glass $\text{Li}_{x7}\text{Si}_{y7}\text{S}_{z3}$, and P_2S_5 type glass $\text{Li}_{x8}\text{P}_{y8}\text{S}_{z4}$, where $0 < x_1 < 2$, $0 < y_1 < 3$, $0 < x_2 < 2$, $0 < y_2 < 1$, $0 < z_1 < 3$, $0 < x_3 < 4$, $0 < 3 < 13$, $0 < x_4 < 2$, $0 < y_4 < 3$, $0 < x_5 < 4$, $0 < y_5 < 1$, $0 < z_2 < 1$, $0 < w < 5$, $0 < x_6 < 4$, $0 < y_6 < 2$, $0 < x_7 < 3$, $0 < y_7 < 2$, $0 < z_3 < 4$, $0 < x_8 < 3$, $0 < y_8 < 3$, and $0 < z_4 < 7$. Thus, the ionic conductivity of the separator can be further increased.

[0086] Optionally, the inorganic particles capable of having electrochemical reactions may include at least one of lithium-containing transition metal oxide, lithium-containing phosphate, a carbon-based material, a silicon-based material, a tin-based material, and a lithium titanium compound.

[0087] Optionally, the organic particles may include one or more of polycarbonate, polythiophene, polypyridine, polystyrene, polyacrylic wax, polyethylene, polypropylene, cellulose, cellulose modifier (such as carboxymethyl cellulose), melamine resin, phenolic resin, polyester (such as polyethylene terephthalate, polyethylene naphthalate, and polybutylene terephthalate), organosilicon resin, polyimide, polyamide-imide, polyarylamide, polyphenylene sulfide,

polysulfone, polyether sulfone, polyether ether ketone, polyarylether ketone, and a copolymer of butyl acrylate and ethyl methacrylate (such as a crosslinked polymer of butyl acrylate and ethyl methacrylate).

[0088] Optionally, the organic-metal framework material may include one or more of a nitrogen-containing heterocyclic ligand structure, an organic carboxylic acid ligand structure, and a nitrogen-oxygen mixed ligand structure.

[0089] In some embodiments, a percentage of the binder may be greater than or equal to 10%, optionally 10%-30%, based on a total weight of the binding layer.

[0090] In some embodiments, a percentage of the filler may be less than or equal to 90%, optionally 40%-90% or 60%-80%, based on the total weight of the binding layer.

[0091] In some embodiments, the binding layer may further include a dispersant such as carboxymethyl cellulose, to adjust the viscosity of a binding layer slurry and improving the quality and uniformity of the binding layer.

[0092] In some embodiments, a percentage of the dispersant may be less than or equal to 25%, optionally less than or equal to 20%, based on the total weight of the binding layer.

[0093] The thickness of the binding layer is greater than or equal to 0.3 μm , optionally 0.5-4 μm .

[0094] According to some embodiments, the thickness of the binding layer may be tested using devices and methods known in the art. Specifically, a scanning electron microscope (such as ZEISS Sigma 300) is used to obtain a cross-sectional scanning electron microscope (SEM) image of the separator with reference to JY/T010-1996. In an example, the test may be performed according to the following method: a plurality of regions are randomly selected from a cross section of the separator, the thickness of a coating is measured at a specified magnification for at least five times, measured values of different regions are statistically recorded, and a mean value is taken as the thickness of the binding layer.

[0095] In any embodiment of this application, the air permeability of the separator is less than or equal to 400 s/100 cc, optionally 250 s/100 cc-320 s/100 cc; the transverse direction tensile strength of the separator is 1500-4500 kgf/cm.^{sup.2}, optionally 3000-4500 kgf/cm.^{sup.2}; the longitudinal direction tensile strength of the separator is 1500-4500 kgf/cm.^{sup.2}, optionally 3000-4500 kgf/cm.^{sup.2}; the transverse direction thermal shrinkage rate of the separator at 250° C. for 1 h is 0-0.4%, optionally 0%-0.2%; and the longitudinal direction thermal shrinkage rate of the separator at 250° C. for 1 h is 0-0.4%, optionally 0%-0.2%. For example, in one embodiment, the air permeability of the separator is optionally 260 s/100 cc; the transverse direction thermal shrinkage rate of the separator at 250° C. for 1 h is optionally 0.4%; and the longitudinal direction thermal shrinkage rate of the separator at 250° C. for 1 h is optionally 0.15%. When at least one of the above properties of the separator meets the given range, the separator has good heat resistance and physical properties, thereby improving the reliability of the secondary battery.

[0096] According to some embodiments, the air permeability, the transverse direction tensile strength, longitudinal direction tensile strength, transverse direction thermal shrinkage rate, and longitudinal direction thermal shrinkage rate of the first base film and the second base film all have meanings well known in the art and can be tested using devices and methods known in the art. For example, the test may be performed in accordance with the standard GB/T 36363-2018.

[Preparation Method of Separator]

[0097] An embodiment of this application further provides a method for preparing the separator provided by the embodiments of this application. The preparation method includes: providing a first base film and a second base film, and laminating the first base film and the second base film to obtain the separator according to any one of the foregoing embodiments. A melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film.

[0098] Referring to FIG. 3, an embodiment of this application provides a preparation method of separator. The preparation method includes the following steps.

[0099] **S100.** Provide a first base film and a second base film, where a melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film.

[0100] **S200.** Provide a binding layer slurry, where the binding layer slurry includes a binder. Optionally, the binding layer slurry includes the binder and a filler. The provision of the binding layer between the first base film and the second base film can not only compensate process defects in an over-hot-pressing lamination process, but also further improve the heat resistance and physical properties of the separator, thereby improving the reliability of the secondary battery.

[0101] **S300.** Coat the first base film and/or the second base film with the binding layer slurry to form a binding layer. For example, the binding layer slurry may be applied onto only one surface of the first base film or the second base film; and the binding layer slurry may be applied onto both the first base film and the second base film to form the binding layer.

[0102] **Step S400.** Laminate the first base film and the second base film to obtain the separator according to any one of the foregoing embodiments. The first base film and the second base film may be laminated through hot pressing. In the hot-pressing lamination process, an excessively high temperature results in low porosity and poor air permeability, and an excessively low temperature results in insufficient adhesion between the first base film and the second base film. Therefore, it is necessary to provide an appropriate hot pressing temperature through adjustment. Optionally, the hot pressing temperature is 20° C.-50° C.

[0103] Unless otherwise specified, raw materials (such as the first base film, the second base film, the binder, the filler, and the like) used in the preparation method of the separator are all commercially available.

[Positive Electrode Plate]

[0104] In a secondary battery, a positive electrode plate generally includes a positive electrode current collector and a positive electrode film layer disposed on the positive electrode current collector, where the positive electrode film layer includes a positive electrode active material.

[0105] The positive electrode current collector may be a common metal foil sheet or a composite current collector (the composite current collector may be formed by providing a metal material on a polymer matrix). In an example, an aluminum foil may be used as the positive electrode current collector.

[0106] A specific type of the positive electrode active material is not limited. Active materials known in the art that can be used for positive electrodes of secondary batteries can be used, and persons skilled in the art can select an active material based on actual needs.

[0107] In an example, the positive electrode active material may include but is not limited to one or more of lithium transition metal oxide, olivine-structured lithium-containing phosphate, and respective modified compounds thereof. An Example of the lithium transition metal oxide may include but is not limited to one or more of lithium cobalt oxide, lithium nickel oxide, lithium manganese oxide, lithium nickel cobalt oxide, lithium manganese cobalt oxide, lithium nickel manganese oxide, lithium nickel cobalt manganese oxide, lithium nickel cobalt aluminum oxide, and modified compounds thereof. An example of the olivine-structured lithium-containing phosphate may include but is not limited to one or more of lithium iron phosphate, a composite material of lithium iron phosphate and carbon, lithium manganese phosphate, a composite material of lithium manganese phosphate and carbon, lithium manganese iron phosphate, a composite material of lithium manganese iron phosphate and carbon, and modified compounds thereof. These materials are all commercially available.

[0108] The modified compounds of the foregoing materials may be obtained through doping modification and/or surface coating modification to the materials.

[0109] Generally, the positive electrode film layer optionally further includes a binder, a conductive agent, and another optional additive.

[0110] In an example, the conductive agent may be one or more of superconducting carbon,

acetylene black, carbon black, Ketjen black, carbon dots, carbon nanotubes, conductive carbon black (Super P, SP), graphene, and carbon nanofiber.

[0111] In an example, the binder may be one or more of polymerized styrene butadiene rubber (Polymerized Styrene Butadiene Rubber, SBR), water-based acrylic resin (water-based acrylic resin), polyvinylidene difluoride (Polyvinylidene Difluoride, PVDF), Polytetrafluoroethylene (Polytetrafluoroethylene, PTFE), ethylene vinyl acetate copolymer (ethylene vinyl Acetate Copolymer, EVA), polyacrylic acid (Polyacrylic Acid, PAA), carboxymethyl cellulose (Carboxymethyl Cellulose, CMC), polyvinyl alcohol (Polyvinyl Alcohol, PVA), and polyvinyl butyral (Polyvinyl Butyral, PVB).

[Negative Electrode Plate]

[0112] In a secondary battery, a negative electrode plate generally includes a negative electrode current collector and a negative electrode film layer disposed on the negative electrode current collector, where the negative electrode film layer includes a negative electrode active material.

[0113] The negative electrode current collector may be a common metal foil sheet or a composite current collector (for example, the composite current collector may be formed by providing a metal material on a polymer matrix). In an example, a copper foil may be used as the negative electrode current collector.

[0114] A specific type of the negative electrode active material is not limited. Active materials known in the art that can be used for negative electrodes of secondary batteries can be used, and persons skilled in the art can select an active material based on actual needs. In an example, the negative electrode active material may include but is not limited to one or more of artificial graphite, natural graphite, hard carbon, soft carbon, a silicon-based material, and a tin-based material. The silicon-based material may be selected from one or more of elemental silicon, a silicon-oxygen compound (such as silicon monoxide), a silicon-carbon compound, a silicon-nitrogen compound, and a silicon alloy. The tin-based material may be selected from one or more of elemental tin, a tin-oxygen compound, and a tin alloy. These materials are all commercially available.

[0115] In some embodiments, the negative electrode active material may include a silicon-based material to further increase the energy density of the battery.

[0116] Generally, the negative electrode film layer optionally further includes a binder, a conductive agent, and another optional additive.

[0117] In an example, the conductive agent may be one or more of superconducting carbon, acetylene black, carbon black, Ketjen black, carbon dots, carbon nanotubes, graphene, and carbon nanofiber.

[0118] In an example, the binder may be one or more of styrene butadiene rubber (SBR), water-based acrylic resin (water-based acrylic resin), polyvinylidene difluoride (PVDF), polytetrafluoroethylene (PTFE), ethylene vinyl acetate copolymer (EVA), polyvinyl alcohol (PVA), and polyvinyl butyral (PVB).

[0119] In an example, the another optional additive may be a thickening and dispersing agent (for example, sodium carboxymethyl cellulose, CMC-Na) or a PTC thermistor material.

[Electrolyte]

[0120] The secondary battery may include an electrolyte, and the electrolyte conducts ions between a positive electrode and a negative electrode. The electrolyte may include an electrolytic salt and a solvent.

[0121] In an example, the electrolytic salt may be selected from one or more of lithium hexafluorophosphate (LiPF₆), lithium tetrafluoroborate (LiBF₄), lithium perchlorate (LiClO₄), lithium hexafluoroarsenate (LiAsF₆), lithium bis(fluorosulfonyl)imide (LiFSI), lithium bistrifluoromethanesulfonimide (LiTFSI), lithium trifluoromethanesulfonate (LiTFS), lithium difluoro (oxalato) borate (LiDFOB), lithium dioxalate borate (LiBOB), lithium difluorophosphate (LiPO₂F₂), lithium difluoro (dioxalato)phosphate (LiDFOP), and

lithium tetrafluoro oxalato phosphate (LiTFOF).

[0122] In an example, the solvent may be selected from one or more of ethylene carbonate (Ethylene Carbonate, EC), propylene carbonate (Propylene Carbonate, PC), ethyl methyl carbonate (Ethyl Methyl Carbonate, EMC), diethyl carbonate (Diethyl Carbonate, DEC), dimethyl carbonate (Dimethyl Carbonate, DMC), dipropyl carbonate (Dipropyl Carbonate, DPC), methyl propyl carbonate (Methyl Propyl Carbonate, MPC), ethyl propyl carbonate (Ethyl Propyl Carbonate, EPC), butylene carbonate (Butylene Carbonate, BC), fluoroethylene carbonate (Fluoroethylene Carbonate, FEC), methyl formate (Methyl Formate, MF), methyl acetate (Methyl Acetate, MA), ethyl acetate (Ethyl Acetate, EA), n-propyl acetate (n-Propyl Acetate, PA), methyl propionate (Methyl Propionate, MP), ethyl propanoate (Ethyl Propanoate, EP), n-propyl propionate (n-Propyl Propionate, PP), methyl butyrate (Methyl Butyrate, MB), ethyl butyrate (Ethyl Butyrate, EB), 1,4-butyrolactone (1,4-Butyrolactone, GBL), tetramethylene sulfone (Tetramethylene Sulfone, SF), methyl sulfone (Methyl Sulfone, MSM), methyl ethyl sulfone (Methyl Ethyl Sulfone, EMS), and diethyl sulfone (Diethyl Sulfone, ESE).

[0123] In some embodiments, the electrolyte further includes an additive. For example, the additive may include a negative electrode film-forming additive, or may include a positive electrode film-forming additive, or may include an additive capable of improving some performance of the battery, for example, an additive for improving overcharge resistance performance of the battery, an additive for improving high-temperature performance of the battery, or an additive for improving low-temperature performance of the battery.

[0124] In some embodiments, the secondary battery may be a lithium-ion secondary battery.

[0125] In any embodiment of this application, the second base film of the separator faces the negative electrode plate. The second base film faces the negative electrode plate, so that the second base film facing the negative electrode plate can complement the swelling resistance, the separator can have more stable physical properties, and in a long-term use environment of the battery, even if the first base film loses the insulation performance due to swelling, the second base film can complement the performance, preventing short circuits between electrodes and also guaranteeing the strength of the separator, thereby improving the reliability and cycle life of the battery in long-term cycle life.

[0126] The two layers of base film structures, that are, the first base film and the second base film, cooperate with each other, and the swelling rate of the first base film is smaller than the swelling rate of the second base film, so that in a long-term use environment of the separator, even if one layer of base film loses the insulation performance due to swelling, the other layer of base film can complement the performance, reducing the risk of short circuits between electrodes and also enhancing the strength of the separator, thereby improving the reliability of the battery in long-term service life.

[0127] The secondary battery is not particularly limited in shape in the embodiments of this application and may be cylindrical, rectangular, or of any other shapes. FIG. 4 shows a secondary battery 5 of a rectangular structure as an example.

[0128] In some embodiments, the secondary battery may include an outer package. The outer package is used for packaging the positive electrode plate, the negative electrode plate, and the electrolyte. The second base film of the separator faces the negative electrode plate.

[0129] In some embodiments, the outer package of the secondary battery may be a hard shell, for example, a hard plastic shell, an aluminum shell, or a steel shell. The outer package of the secondary battery may alternatively be a soft pack, for example, a soft pouch. A material of the soft package may be plastic, for example, may include one or more of polypropylene (polypropylene, PP), polybutylene terephthalate (polybutylene terephthalate, PBT), and polybutylene succinate (polybutylene succinate, PBS).

[0130] In some embodiments, referring to FIG. 5, the outer package may include a housing body 51 and a cover plate 53. The housing body 51 may include a base plate and a side plate connected

onto the base plate, and the base plate and the side plate enclose an accommodating cavity. The housing body **51** has an opening in communication with the accommodating cavity, and the cover plate **53** can cover the opening to close the accommodating cavity.

[0131] The preparation method of the secondary battery in this application is well known. In some embodiments, a positive electrode plate, a separator, a negative electrode plate, and an electrolyte may be assembled into a secondary battery. For example, the positive electrode plate, separator, and negative electrode plate may be made into an electrode assembly through winding and/or lamination; and the electrode assembly is placed into an outer package, followed by drying, and the electrolyte is injected, followed by processes such as vacuum packaging, standing, formation, and shaping, to obtain a battery cell. A plurality of battery cells may be further connected in series, parallel, or series-parallel to form a battery module. A plurality of battery modules may be further connected in series, parallel, or series-parallel to form a battery pack. In some embodiments, the plurality of battery cells may alternatively directly form a battery pack.

[0132] FIG. **6** shows a battery module **4** as an example. Referring to FIG. **6**, in the battery module **4**, a plurality of secondary batteries **5** may be sequentially arranged in a length direction of the battery module **4**. Certainly, the secondary batteries may alternatively be arranged in any other manners. Further, the plurality of secondary batteries **5** may be fastened using fasteners.

[0133] The battery module **4** may alternatively include a housing with an accommodating space, and the plurality of secondary batteries **5** are accommodated in the accommodating space.

[0134] In some embodiments, the battery module may be further assembled into a battery pack, and a quantity of battery modules included in the battery pack may be adjusted based on application and capacity of the battery pack.

[0135] FIG. **7** and FIG. **8** show a battery pack **1** as an example. Referring to FIG. **7** and FIG. **8**, the battery pack **1** may include a battery box and a plurality of battery modules **4** arranged in the battery box. The battery box includes an upper box body **2** and a lower box body **3**. The upper box body **2** can cover the lower box body **3** to form an enclosed space for accommodating the battery modules **4**. The plurality of battery modules **4** may be arranged in the battery box in any manner.

[0136] In any embodiment of this application, the second base film of the separator faces the positive electrode plate. The second base film has good swelling resistance and faces the positive electrode, and thus can reduce the possibility of thermal runaway, thereby improving the reliability of the secondary battery.

[Apparatus]

[0137] This application further provides an electric apparatus, where the electric apparatus includes the secondary battery of this application. The battery cell, battery module, or battery pack can be used as a power source for the apparatus, or as an energy storage unit for the apparatus. The apparatus may be but is not limited to a mobile device (for example, a mobile phone or a notebook computer), an electric vehicle (for example, a battery electric vehicle, a hybrid electric vehicle, a plug-in hybrid electric vehicle, an electric bicycle, an electric scooter, an electric golf vehicle, or an electric truck), an electric train, a ship, a satellite, or an energy storage system.

[0138] A battery cell, battery module, or battery pack may be selected according to requirements for using the electric apparatus.

[0139] FIG. **9** shows an electric apparatus as an example. The electric apparatus may be a battery electric vehicle, a hybrid electric vehicle, or a plug-in hybrid electric vehicle. To satisfy requirements of the electric apparatus for high power and high energy density of the secondary battery, a battery pack or a battery module may be used.

[0140] In another example, the electric apparatus may be a mobile phone, a tablet computer, or a notebook computer. Such electric apparatus is generally required to be light and thin and may use a battery cell as its power source.

[0141] The following further describes beneficial effects of this application with reference to examples.

[0142] To describe the technical problems solved by the embodiments of this application, technical solutions, and beneficial effects more clearly, the following further provides descriptions in detail with reference to the embodiments and accompanying drawings. Apparently, the described embodiments are only some but not all of the embodiments of this application. The following description of at least one exemplary embodiment is merely illustrative and definitely is not construed as any limitation on this application or on use of this application. All other embodiments obtained by persons of ordinary skill in the art based on the embodiments of this application without creative efforts shall fall within the protection scope of this application.

I. Preparation of the Battery

Example 1

1. Preparation of Positive Electrode Plate

[0143] A positive electrode active material $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_{2}$ (NCM811), a conductive agent carbon black (Super P), and a binder polyvinylidene difluoride (PVDF) at a mass ratio of 96.2:2.7:1.1 were mixed to uniformity in an appropriate amount of solvent N-methylpyrrolidone (N-Methylpyrrolidone, NMP) to obtain a positive electrode slurry. The positive electrode slurry was evenly applied onto a positive electrode current collector aluminum foil, followed by drying, cold pressing, slitting, and cutting, to obtain a positive electrode plate.

2. Preparation of Negative Electrode Plate

[0144] A negative electrode active material artificial graphite, a conductive agent carbon black (Super P), a binder styrene butadiene rubber (SBR), and sodium carboxymethyl cellulose (CMC-Na) at a mass ratio of 96.4:0.7:1.8:1.1 were mixed to uniformity in an appropriate amount of solvent deionized water to form a negative electrode slurry. The negative electrode slurry was evenly applied onto a negative electrode current collector copper foil, followed by drying, cold pressing, slitting, and cutting, to obtain a negative electrode plate.

3. Preparation of Separator

[0145] (1) A first base film was provided, which was made of a material including polypropylene (PP), with a melting point of 165°C . and a swelling rate of 0.01%.

[0146] (2) A second base film was provided, which was made of a material including polyethylene (PE), with a melting point of 135°C . and a swelling rate of 8%.

[0147] (3) Preparation of a binding layer slurry: boehmite, polyacrylate, and carboxymethyl cellulose at a ratio of 4:1:1 were mixed to uniformity in an appropriate amount of solvent deionized water, to form the binding layer slurry.

[0148] (4) The binding layer slurry obtained in step (3) was applied onto the PE base film to form a binding layer, and the PP and PE base films were subjected to hot-pressing molding to ensure that the binding layer was between the PP base film and the PE base film, to prepare a separator.

(4) Preparation of Electrolyte

[0149] Ethylene carbonate (EC) and ethyl methyl carbonate (EMC) at a mass ratio of 30:70 were mixed to obtain an organic solvent. A fully dried electrolytic salt LiPF_6 was dissolved in the solvent mixture, with a concentration of the electrolytic salt being 1.0 mol/L; and then the electrolytic salt and the solvent mixture were mixed to uniformity to obtain an electrolyte.

5. Preparation of Secondary Battery

[0150] The positive electrode plate, separator, and negative electrode plate were stacked in sequence to ensure that the separator is located between the positive electrode plate and the negative electrode plate for separation; then, the stack was wound to obtain an electrode assembly; the electrode assembly was placed into an outer package; and the prepared electrolyte was injected into a dried housing body, followed by processes such as vacuum packaging, standing, formation, and shaping to obtain a secondary battery.

[0151] The preparation methods of the secondary batteries in Examples 2 to 17 and Comparative examples 1 to 6 were similar to that of the secondary battery in Example 1, except that different separators were used. Specific details are shown in Table 1.

II. Performance Test

1. Thermal Shrinkage Rate of the Separator (250° C.)

[0152] Sample preparation: the above prepared separator was cut into samples with a width of 50 mm and a length of 100 mm; five parallel samples were placed on a piece of A4 paper; and then the A4 paper with the samples was placed on a piece of corrugated paper with a thickness of 1 mm to 5 mm.

[0153] Sample testing: the A4 paper on the corrugated paper was put into a blowing-type oven; the blowing-type oven was set at a temperature of 250° C.; after the temperature reached a set value and stabilizes for 30 minutes, timing was started; and after a set time (1 hour for this application) was reached, a width of the separator was measured and recorded as a.

[0154] Transverse direction (TD) thermal shrinkage rate= $[(50-a)/50] \times 100\%$. An average value of the 5 parallel samples was taken as a test result.

2. Battery Cycle Life

[0155] At 25° C., the batteries in the above examples and comparative examples were charged to 3.65 V at a constant current of 1C, then charged to a current $\leq 0.05C$ at a constant voltage of 3.65 V, and then discharged to a voltage of 2.5 V at a constant current of 1C. This was one charge-discharge cycle, and a discharge capacity at that moment was recorded as a discharge capacity of the first cycle of the battery. The charge-discharge cycle was performed repeatedly, and the number of cycles when the capacity dropped to 80% was recorded.

TABLE-US-00001 TABLE 1 Battery performance test results of examples and comparative examples

Battery performance	Transverse	Separator structure	direction	First base film	Second base film	thermal Melting	Melting Swelling	shrinkage Cycle point	Swelling point	Swelling rate	rate life
Serial No.	Material (° C.)	rate	Material (° C.)	rate	ratio (250° C.)	(number)	Example	1	PP	165	0.01%
0.01%	PE 135	8.0%	0.0013	0.30%	3287	Example 2	PP	165	0.20%	PE 135	8.0%
0.0250	0.15%	3400	Example 3	PP	165	0.50%	PE 135	8.0%	0.0625	0.14%	3392
Example 4	PP	165	1.00%	PE 135	8.0%	0.1250	0.14%	3357	Example 5	PP	165
2.00%	PE 135	8.0%	0.2500	0.15%	3316	Example 6	PP	165	2.50%	PE 135	8.0%
0.3125	0.26%	3200	Example 7	PP	165	5.00%	PE 135	8.0%	0.6250	0.28%	3172
Example 8	PP	165	7.00%	PE 135	8.0%	0.8750	0.34%	3083	Example 9	PP	165
0.20%	PE 135	0.5%	0.4000	0.21%	3146	Example 10	PP	165	0.20%	PE 135	1.0%
0.2000	0.15%	3412	Example 11	PP	165	0.20%	PE 135	1.5%	0.1333	0.16%	3351
Example 12	PP	165	0.20%	PE 135	2.0%	0.1000	0.14%	3371	Example 13	PP	165
0.20%	PE 135	5.0%	0.0400	0.15%	3329	Example 14	PP	165	0.20%	PE 135	10.0%
0.0200	0.12%	3245	Example 15	PP	165	0.20%	PE 135	12.0%	0.0167	0.39%	3042
Example 16	PTFE	327	0.50%	PE 135	8.0%	0.0625	0.16%	3265	Example 17	PET	270
0.70%	PE 135	8.0%	0.0875	0.14%	3261	Comparative	PP	135	5.00%	PE 135	2.70%
1.8519	0.50%	2800	example 1	Comparative	PP	135	2.70%	PE 135	2.70%	1.0000	0.52%
2861	example 2	Comparative	PTFE	327	0.50%	PE 135	0.30%	1.6667	0.43%	3000	example 3
Comparative	PTFE	327	0.50%	PE 135	0.5%	1.0000	0.41%	2912	example 4	Comparative	PET
270	0.70%	PE 135	0.5%	1.4000	0.47%	3006	example 5	Comparative	PET	270	0.70%
PE 135	0.7%	1.0000	0.48%	2939	example 6						

[0156] As can be seen from Table 1, in Example 1 to Example 17, the melting point of the first base film is higher than the melting point of the second base film, and the swelling rate of the first base film is smaller than the swelling rate of the second base film, so that the prepared batteries have longer cycle life. In contrary, the separators used in Comparative example 1 to Comparative example 6 do not meet the design requirements of this application, and batteries prepared using the separators have unsatisfactory cycle life. Therefore, batteries prepared using the separator defined in this application have higher reliability.

[0157] The foregoing descriptions are merely specific embodiments of this application, but are not intended to limit the protection scope of this application. Any equivalent modifications or replacements readily figured out by persons skilled in the art within the technical scope disclosed in

this application shall fall within the protection scope of this application. Therefore, the protection scope of this application shall be subject to the protection scope of the claims.

Claims

1. A separator, comprising: a first base film, and a second base film, wherein a melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film.
2. The separator according to claim 1, wherein a ratio of the swelling rate of the first base film and the swelling rate of the second base film is less than or equal to 0.8, optionally less than or equal to 0.3.
3. The separator according to claim 1, wherein the swelling rate of the first base film is less than or equal to 5%; and/or the swelling rate of the second base film is less than or equal to 10%.
4. The separator according to claim 1, wherein a crystallinity of the first base film is smaller than a crystallinity of the second base film.
5. The separator according to claim 1, wherein a ratio of the crystallinity of the first base film and the crystallinity of the second base film is 0.4-0.85.
6. The separator according to claim 1, wherein the crystallinity of the first base film is 40%-60%; and/or the crystallinity of the second base film is 45%-85%.
7. The separator according to claim 1, wherein a branching degree of the first base film is larger than a branching degree of the second base film.
8. The separator according to claim 1, wherein the branching degree of the first base film is 0.4-0.9; and/or the branching degree of the second base film is 0.01-0.4.
9. The separator according to claim 1, wherein the melting point of the first base film is 155° C.-365° C.; and/or the melting point of the second base film is 130° C.-250° C.
10. The separator according to claim 1, wherein the first base film and the second base film are each independently selected from at least one of polyolefin and a derivative thereof, halogenated polyolefin and a derivative thereof, polyether and a derivative thereof, polyether ether ketone and a derivative thereof, polyester and a derivative thereof, polyimide and a derivative thereof, polyvinyl alcohol and a derivative thereof, polytetrafluoroethylene and a derivative thereof, polyvinyl fluoride and a derivative thereof, polyvinylidene fluoride and a derivative thereof, and polyethylene terephthalate and a derivative thereof.
11. The separator according to claim 1, wherein a binding layer is further disposed between the first base film and the second base film; the binding layer comprises a binder; and optionally, the binding layer comprises the binder and a filler.
12. The separator according to claim 11, wherein the binder comprises one or more of polyacrylate, polyacrylic acid, polytetrafluoroethylene, polyvinylidene fluoride, vinylidene fluoride-trichloroethylene copolymer, polyvinylpyrrolidone, polyvinyl acetate, ethylene-vinyl acetate copolymer, polyethylene oxide, polyarylate, carboxymethyl cellulose, hydroxypropyl cellulose, regenerated cellulose, cellulose acetate, cellulose acetate propionate, cellulose acetate butyrate, polyacrylonitrile, polyvinyl alcohol, polyethylene, polypropylene, starch, and cyanoethyl branched starch; and/or the filler comprises at least one of inorganic particles, organic particles, and an organic-metal framework material.
13. The separator according to claim 1, wherein an air permeability of the separator is less than or equal to 400 s/100 cc; and/or a transverse direction tensile strength of the separator is 1500 kgf/cm.^{sup.2}-4500 kgf/cm.^{sup.2}; a longitudinal direction tensile strength of the separator is 1500 kgf/cm.^{sup.2}-4500 kgf/cm.^{sup.2}; a transverse direction thermal shrinkage rate of the separator at 250° C. for 1 h is less than or equal to 0.4%; and/or a longitudinal direction thermal shrinkage rate of the separator at 250° C. for 1 h is less than or equal to 0.4%.
14. A preparation method of separator, comprising: providing a first base film and a second base

film, wherein a melting point of the second base film is lower than a melting point of the first base film, and a swelling rate of the first base film is smaller than a swelling rate of the second base film; and laminating the first base film and the second base film to obtain the separator.

15. The preparation method according to claim 14, wherein the preparation method further comprises: providing a binding layer slurry, wherein the binding layer slurry comprises a binder; and coating the first base film and/or the second base film with the binding layer slurry to form a binding layer; wherein optionally, the binding layer slurry further comprises a filler; and optionally, the filler comprises at least one of inorganic particles, organic particles, and an organic-metal framework material.

16. A secondary battery, comprising the separator according to claim 1.

17. The secondary battery according to claim 16, wherein the secondary battery further comprises a positive electrode plate and a negative electrode plate, wherein the separator is disposed between the positive electrode plate and the negative electrode plate, and the second base film faces the negative electrode plate.

18. An electric apparatus, comprising the secondary battery according to claim 16.
